penetration of light than the profundal zone.

7.2.11 Ecosystem services (Benefits)

Ecosystem services are defined as the benefits that people derive from nature. Robert Constanza et al (1927) stated "Ecosystem services are the benefits provided to human, through the transformation of resources (or Environmental assets including land, water, vegetation and atmosphere) into a flow of essential goods and services".

Study on ecosystem services acts as an effective tool for gaining knowledge on ecosystem benefits and their sustained use. Without such knowledge gain, the fate of any ecosystem will be at stake and the benefits they provide to us in future will become bleak.



Robert Constanza and his colleagues estimated the value of global ecosystem services based on various parameters. According to them in 1997, the average global value of ecosystems services estimated was US $ 33 trillion a year. The updated estimate for the total global ecosystem services in 2011 is US $ 125 trillion / year, indicating a four-fold increase in ecosystem services from 1997 to 2011.

Mangrove ecosystem services



- Offers habitat and act as nursery for aquatic plants and animals
- Provides medicine, fuel wood and timber.
- Act as bridge between sea and rivers by balancing sedimentation and soil erosion.
- Help to reduce water force during cyclones, tsunamis and high tide periods.
- Help in wind break, O_2 production, carbon sequestration and prevents salt spray from waves.

How do anthropogenic activities affect ecosystem services?

Now, we all exploit the ecosystem more than that of our needs. The **Millennium Ecosystem Assessment (2005)** found that "over the past 50 years, humans have changed the ecosystem more rapidly and extensively than in any comparable period of time in human history, largely to meet rapidly growing demands for food, fresh water, medicine, timber, fiber and fuel."

The varieties of benefits obtained from the ecosystem are generally categorized into the following four types

Ecosystem services

Provisioning services
- Food, fiber and fuel
- Genetic resources
- Bio-chemicals
- Fresh water
- Medicines

Cultural services
- Spiritual and religious values
- Knowledge system
- Education and inspiration
- Recreation and aesthetic values
- Ecotourism

Supporting services
- Primary production
- Provision of habitat
- Nutrient cycling
- Soil formation and retention
- Production of atmospherc oxygen
- Water cycling

Regulating services
- Invasion resistance
- Herbivory pollination
- Seed dispersal
- Climate regulation
- Pest regulation
- Disease regulation
- Erosion regulation
- Water purification
- Natural hazard protection

Figure 7.17: Types of Ecosystem services

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